



Lab Final Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	40
Understanding/Analysis	10					
Implementation	15					
Accuracy	10					
Task Efficiency	5					
Total obtained mark						
Comments						

Semester: Summer 2026

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Batch: 46

Section: J

Course Code: SE132 Course Name: Data Structure Lab

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Designation: Lecturer

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Multi-Stop Bus Reservation System

1. Problem Statement

Manual bus ticket booking is slow, error-prone, and cannot track seat availability in real time. Conventional systems also lock a seat for the whole trip, even when different passengers travel between different stops, wasting capacity. This project builds a Bus Ticket Reservation System that assigns seats automatically, allows one seat to be shared by passengers travelling non-overlapping route segments, and places passengers on a waiting list only when no seat can serve their journey.

2. Benefits

- 1. Increased seat utilization:** By tracking bookings at the level of route segments rather than the entire journey, a single seat can serve multiple passengers.
- 2. Fair passenger management:** A waiting list is maintained strictly on a first-come, first-served basis, avoiding unnecessary or premature rejection of passengers.
- 3. Faster operations:** Seat status, cancellations, and waiting list assignments are processed instantly, replacing slow manual lookups with direct data access.
- 4. Speed and accuracy:** Instant booking, cancellation, and search, with no double-booking errors.
- 5. Scalability:** The underlying design can be extended to additional stops, multiple bus routes, or larger fleets with minimal structural change.

3. Used Data Structure & Reasoning

Array (seats[]): Fixed number of seats, so an array gives $O(1)$ direct access by seat number.

Linked list (per-seat bookings): Each seat can hold a variable number of segment bookings, so a linked list grows dynamically. New nodes are inserted at the head ($O(1)$).

Queue (waiting list): Fairness requires strict first-come, first-served order, which a queue (enqueue at rear, dequeue at front) guarantees in $O(1)$.

Interval overlap check: Two segments $[s1, e1)$ and $[s2, e2)$ conflict only if $s1 < e2$ and $s2 < e1$; this rule decides whether a seat can be shared.

GitHub Link: <https://github.com/tanvirIzBack/Multi-Stop-Bus-Reservation-System.git>

Output:

```
==== Multi-Stop Bus Reservation System ====
```

```
Seat summary:
```

```
Seat 1: 0 booking(s)
```

```
Seat 2: 0 booking(s)
```

```
Seat 3: 0 booking(s)
```

1. Book a Ticket
2. Cancel a Ticket
3. Check Seat Bookings
4. View Waiting List
5. View Route Stops
6. Exit

```
Enter your choice: 1
```

```
Enter passenger name: Elon
```

```
Available stops:
```

```
0. Cox's Bazar
```

```
1. Chittagong
```

```
2. Comilla
```

```
3. Dhaka
```

```
Boarding stop number: 0
```

```
Destination stop number: 1
```

```
Ticket booked for Elon on Seat 1 (Cox's Bazar -> Chittagong).
```

```
Enter your choice: 1
```

```
Enter passenger name: Tanvir
```

```
Available stops:
```

```
0. Cox's Bazar
```

```
1. Chittagong
```

```
2. Comilla
```

```
3. Dhaka
```

```
Boarding stop number: 2
```

```
Destination stop number: 3
```

```
Ticket booked for Tanvir on Seat 1 (Comilla -> Dhaka).
```

==== Multi-Stop Bus Reservation System ====

Seat summary:

Seat 1: 2 booking(s)

Seat 2: 1 booking(s)

Seat 3: 2 booking(s)

1. Book a Ticket
2. Cancel a Ticket
3. Check Seat Bookings
4. View Waiting List
5. View Route Stops
6. Exit

Enter your choice: 3

Enter seat number to check: 3

Seat 3 bookings:

- Urli: Cox's Bazar -> Chittagong
- Kamal: Chittagong -> Dhaka

Enter your choice: 1

Enter passenger name: Roni

Available stops:

0. Cox's Bazar
1. Chittagong
2. Comilla
3. Dhaka

Boarding stop number: 1

Destination stop number: 3

No seat available for that segment. Adding Roni to the waiting list.

Enter your choice: 1

Enter passenger name: Jami

Available stops:

0. Cox's Bazar
1. Chittagong
2. Comilla
3. Dhaka

Boarding stop number: 0

Destination stop number: 3

No seat available for that segment. Adding Jami to the waiting list.

==== Multi-Stop Bus Reservation System ====

Seat summary:

Seat 1: 2 booking(s)

Seat 2: 1 booking(s)

Seat 3: 2 booking(s)

1. Book a Ticket
2. Cancel a Ticket
3. Check Seat Bookings
4. View Waiting List
5. View Route Stops
6. Exit

Enter your choice: 4

--- Waiting List ---

1. Roni (Chittagong -> Dhaka)
2. Jami (Cox's Bazar -> Dhaka)
3. Nehal (Comilla -> Dhaka)

Enter your choice: 2

Enter seat number to cancel from: 3

Current bookings on Seat 3:

- Urli (Cox's Bazar -> Chittagong)
- Kamal (Chittagong -> Dhaka)

Enter passenger name to cancel: Kamal

Booking for Kamal (Chittagong -> Dhaka) on Seat 3 cancelled.

Seat 3 now booked for Roni (Chittagong -> Dhaka) from waiting list.

Enter your choice: 2

Enter seat number to cancel from: 2

Current bookings on Seat 2:

- Johra (Cox's Bazar -> Dhaka)

Enter passenger name to cancel: Johra

Booking for Johra (Cox's Bazar -> Dhaka) on Seat 2 cancelled.

Seat 2 now booked for Jami (Cox's Bazar -> Dhaka) from waiting list.